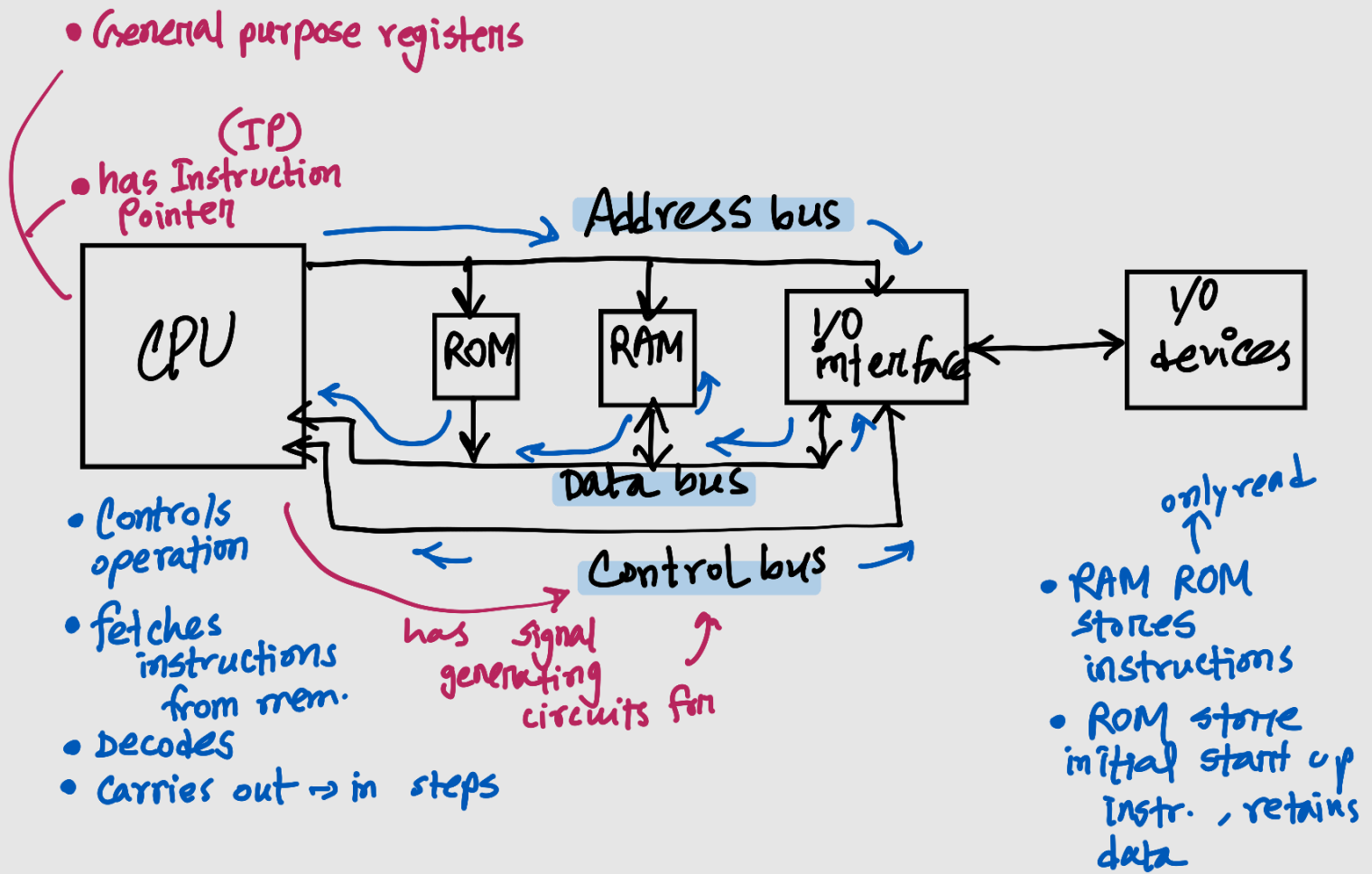


Simple Block Diagram of Microcomputer



• WRITE operation — mem / I/O chip receive data from microprocessor $\rightarrow$ data written into mem location / I/O (register) port

READ operation — mem / I/O chip send data to microprocessor $\rightarrow$ data read from mem location / I/O (register) port

system bus - • Address

- Unidirectional

- Address sent to write to or read from

- Data

- Bidirectional

-

4 bit data
- Nibble

- Control

- Signals used to synchronize operation

- eg. Read, write, Read/not-write, interrupt.

*
Q. How do address locations and memory size have close relation with Address Bus length?

⇒ Suppose a microprocessor has N address lines, each carry 1 bit (0/1)

Possible addressable locations - 2^N unique address

Each address - 1 memory location

If 1 address = 1 byte,

Max memory size = 2^N bytes

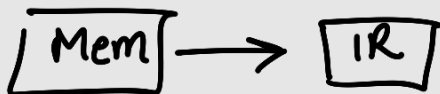
e.g.

$N = 16$ bit, $2^N = 65,536$ address locations, 64 KB memory size

Fetch & Execution Cycle

Fetch cycle

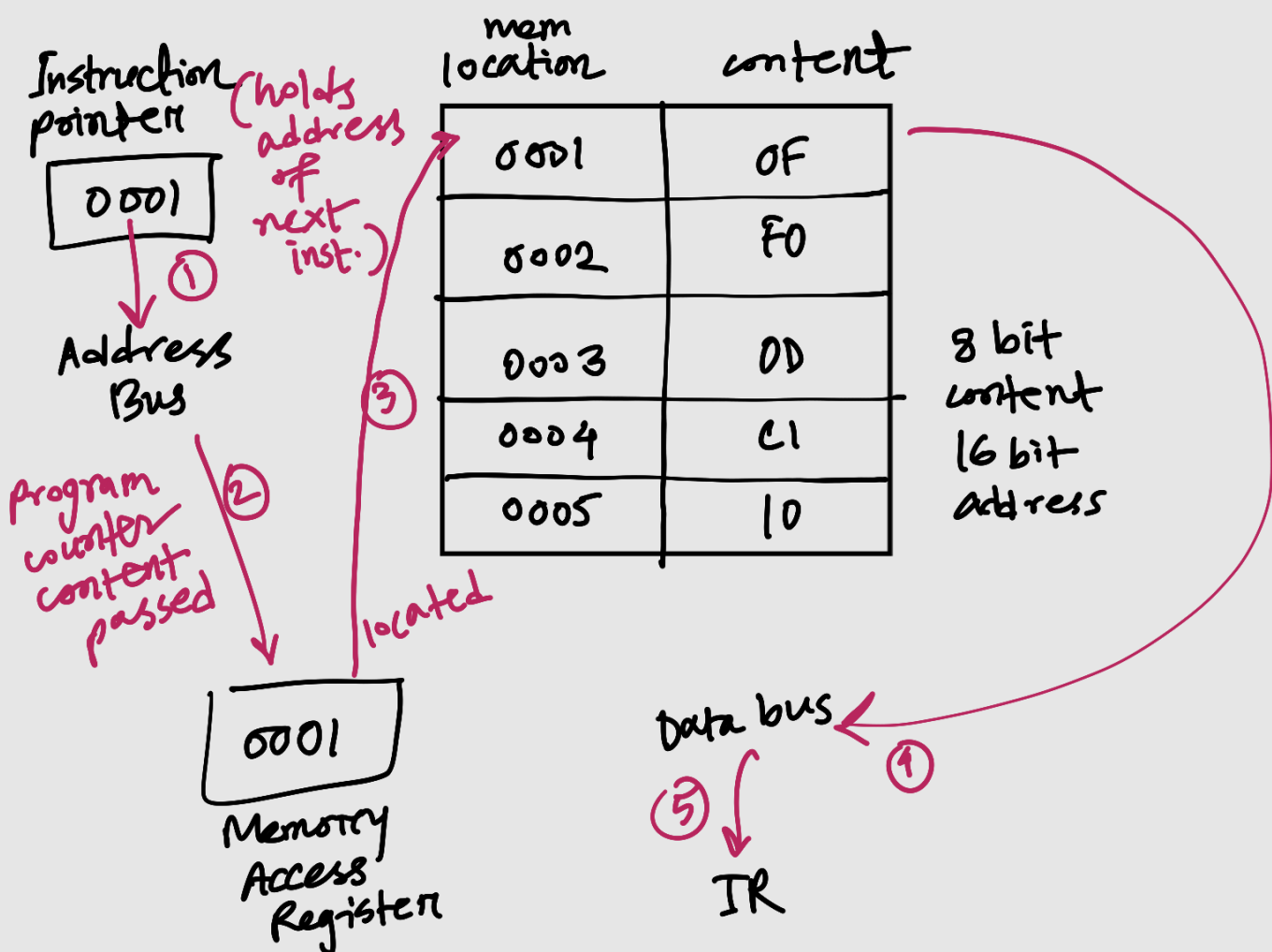
- Takes instruction from mem., stores in instruction register (IR)
- Moves program counter on one so it points to next instr.



Program Counter = 1 (next)

Execute cycle

- Actual actions
- Addressing mode dependent & instr.



Microprocessors : 8085

✓ Address Bus $\Rightarrow$ 16 bit

✓ 8 bit microprocessor

✓ Data Bus $\Rightarrow$ 8 bit

DIP $\Rightarrow$ 40 pin

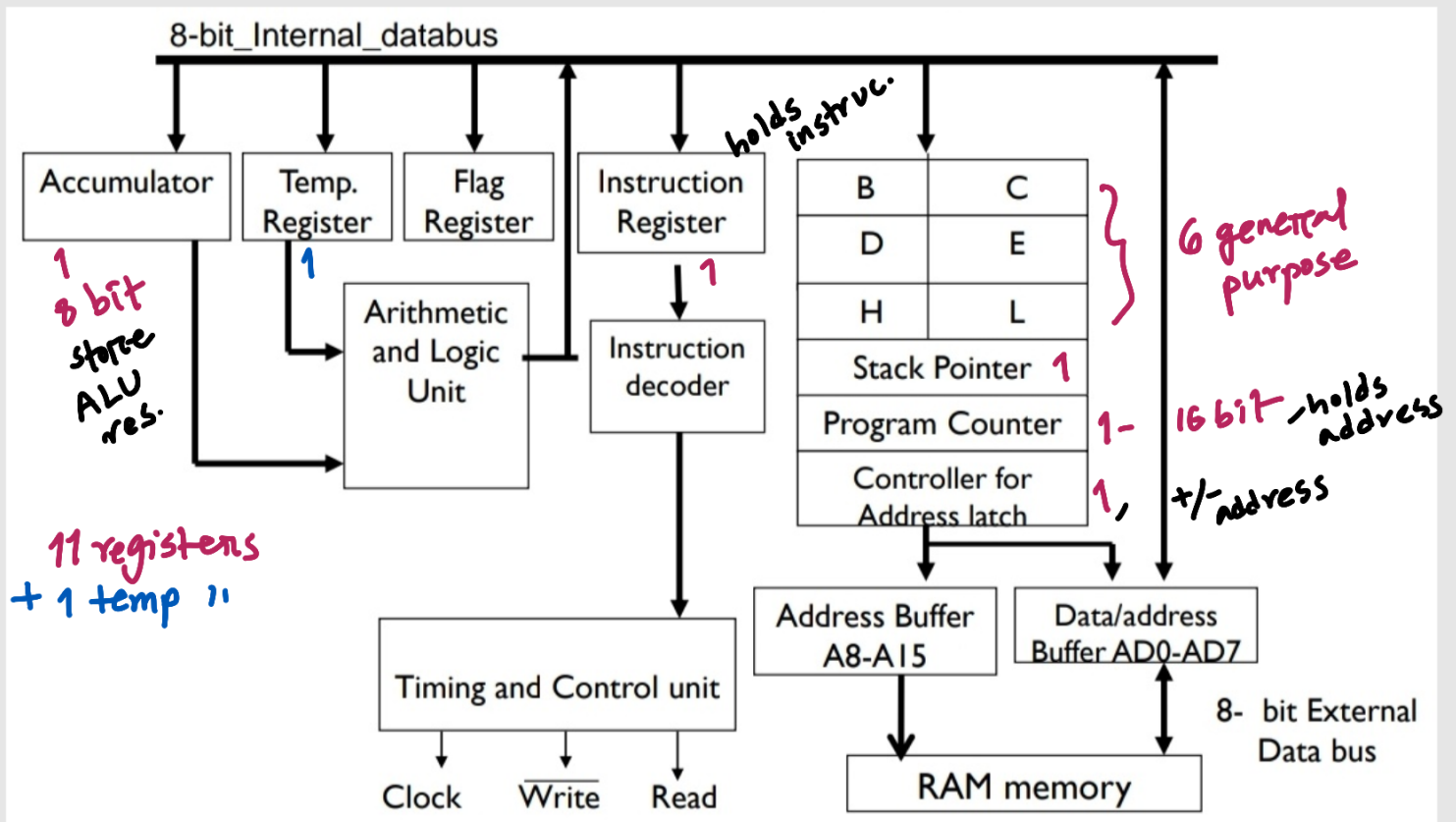
Operation codes $\Rightarrow$ 74

Instructions $\Rightarrow$ 246

✓ 64 KB memory, ✓ 65536 locations

Dual in Line
Package - DIP

Q. Draw schematic architecture of 8-bit 8085 intel processor



Flag register

For latest operation

- sign flag (negative $\rightarrow$ MSB 1, set to 1)
positive $\rightarrow$ MSB 0, reset to 0)
- Zero flag (res 0)
- Auxiliary carry flag (used by system for Binary coded decimal ops.)
- Parity flag (res $\rightarrow$ even 1s, flag = 1
odd 1s, flag = 0 for error checking)
- Carry flag (res exceed 8 bit $\rightarrow$ flag = 1)

* * * Simple Assembly Program in 8085

```
MVI A, 32H
MVI B, 48H
ADD B
OUT 01H
HLT
```

Task:

- Derive the flag values after the ADD instructions

Memory Address	Contents (Binary)	Contents (Hex)	Operation
2000h	0011 1110	3E	Load Reg. Acc.
2001h	0011 0010	32	Value is 32h
2002h	0000 0110	06	Load Reg. B
2003h	0100 1000	48	Value is 48h
2004h	1000 0000	80	Add B with A & Store in A
2005h	1101 0011	D3	Display
2006h	0000 0001	01	Port Id 01h
2007h	0111 1100	76	End

Solution:

$A = 32h \rightarrow 0011\ 0010$ (binary)

$B = 48h \rightarrow 0100\ 1000$ (")

[h $\rightarrow$ hexadecimal]

$$\begin{array}{r} A+B \rightarrow \\ \begin{array}{r} 32\ h \\ 48\ h \\ \hline 7A\ h \end{array} \rightarrow 0111\ 1010 \end{array}$$

sign flag, $S=0$

zero flag, $Z=0$

parity flag, $P=0$ (odd)

carry flag, $CF=0$ (no carry out of MSB) $\rightarrow$ overflow

Auxiliary carry flag, AC

- It is set to 1 when carry from 1st half $\rightarrow$ 2nd half happens during addition

$\begin{array}{cc} 3 & 2 \\ \downarrow & \downarrow \\ \text{upper} & \text{lower} \\ \text{nibble} & \text{nibble} \\ \text{(bits 4-7)} & \text{(bits 0-3)} \\ \text{(2nd half)} & \text{(1st half)} \end{array}$

$$\begin{array}{r} 0010 \\ 1000 \\ \hline 1010 \end{array}$$

In full byte $\rightarrow$
this carries over
to upper nibble

AC=1

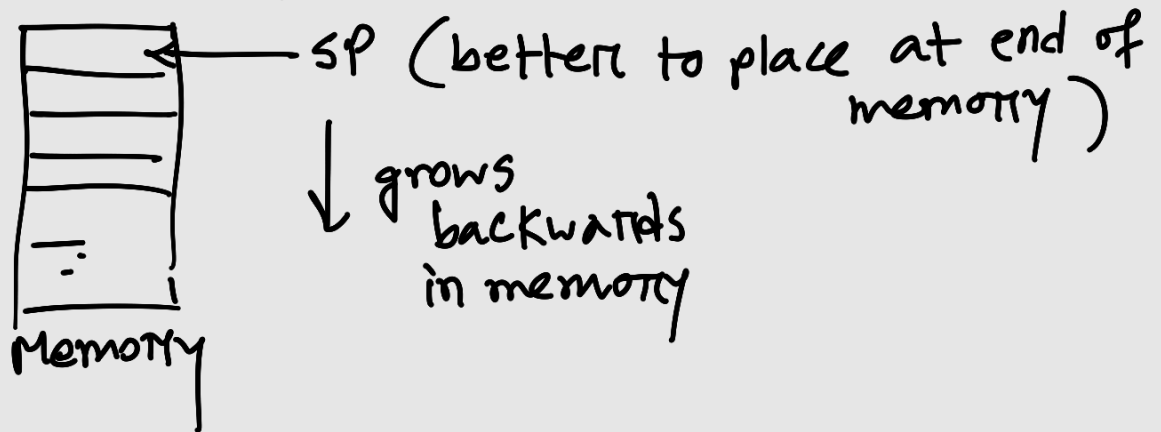
Stack pointer & Stack Memory

- For temporary storage

LIFO

grows backwards in memory

- SP $\rightarrow$ bottom of stack can be defined



LXI SP, FFFFh [Load a 16bit num in a register.]

- Saving info $\rightarrow$ PUSH
Retrieve " $\rightarrow$ POP (only work with register pairs)

*** * *** Q. If $SP = \text{FFFFh}$, show memory locations affected by PUSH B in 8085

Solution -

B register pair - B (high), C (low) [from block diagram]

SP points to the next free location in memory

PUSH B (register pair)

↳ $SP = SP - 1$ (decrement SP)

Copy B (high byte) to 0 (SP memory)

$SP = SP - 1$

Copy C (low byte) to 0 (SP memory)

After operation,

$SP = \text{FFFDh}$ for high byte B, mem = B

$SP = \text{FFFCh}$ for low byte C, mem = C

Finally, $SP = \text{FFFCh}$

Memory locations affected → FFFDh
 FFFCh

Address

FFFDh

FFFCh

Content

B (high byte)

C (low byte)

Q. Show memory contents after POP D

First for low byte E $\rightarrow$ load SP into E

Then high byte D $\rightarrow$ load SP+1 into D

$SP = SP + 2$, initially $SP = \text{FFFC}h$

memory not changed during pop,
only SP moves and registers are loaded

$SP = \text{FFFC}h$

POP D (register pair D+E)

$\hookrightarrow$ Copy SP to E, $E = \text{SP memory}$
 $= \text{FFFC}h$

$SP = SP + 1 = \text{FFFD}h$

Copy SP to D, $D = \text{SP memory}$
 $= \text{FFFD}h$

$SP = SP + 1 = \text{FFFE}h$

<u>Register</u>	<u>Value</u>
D	mem. (FFFDh) (high byte)
E	" (FFFC) (low)
SP	FFFEh